

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

Tony spends \$80 per month on public transportation. A 10-ride pass costs \$12.50, and a single-ride pass costs \$1.50. If g represents the number of 10-ride passes Tony buys in a month and t represents the number of single-ride passes Tony buys in a month, which of the following equations best represents the relationship between g and t ?

Answer

- A. $g + t = 80$
- B. $g + t = 1.50 + 12.50$
- C. $1.50g + 12.50t = 80$
- D. $12.50g + 1.50t = 80$

Correct Answer: D

Rationale

Choice D is correct. Since a 10-ride pass costs \$12.50 and g is the number of 10-ride passes Tony buys in a month, the expression $12.50g$ represents the amount Tony spends on 10-ride passes in a month. Since a single-ride pass costs \$1.50 and t is the number of single-ride passes Tony buys in a month, the expression $1.50t$ represents the amount Tony spends on single-ride passes in a month. Therefore, the sum $12.50g + 1.50t$ represents the amount he spends on the two types of passes in a month. Since Tony spends a total of \$80 on passes in a month, this expression can be set equal to 80, producing $12.50g + 1.50t = 80$.

Choices A and B are incorrect. The expression $g + t$ represents the total number of the two types of passes Tony buys in a month, not the amount Tony spends, which is equal to 80, nor the cost of one of each pass, which is equal to $1.50 + 12.50$. Choice C is incorrect and may result from reversing the cost for each type of pass Tony buys in a month.